

# Mathematical and Computational Methods

## Exercise Sheet 3: Complex numbers II: exponential form, De Moivre, roots

**Tutorial route** (►, in this order, about 80 minutes): 1.3, 2.4, 2.5(a,b), 2.6, 2.8, 2.10(a,b), 2.3. Everything else is for home study.

Levels: ★ drill, ★★ core, ★★ ★ challenge. Give exact answers unless a decimal is asked for.

### 1 Warm-up drills

**Exercise 1.1** (★ drill — exponential form). Write each number in exponential form  $r e^{i\theta}$  with  $r > 0$  and  $-\pi < \theta \leq \pi$ :

(a)  $1 + i$ , (b)  $-3$ , (c)  $2i$ , (d)  $-1 - i$ , (e)  $\sqrt{3} - i$ , (f)  $-2 + 2\sqrt{3}i$ .

**Exercise 1.2** (★ drill — back to Cartesian; Euler's identity). Write each in Cartesian form  $a + ib$ :

(a)  $2 e^{i\pi/3}$ , (b)  $3 e^{i\pi}$ , (c)  $\sqrt{2} e^{-i\pi/4}$ , (d)  $4 e^{i3\pi/2}$ , (e)  $e^{i5\pi/6}$ , (f)  $e^{i2\pi}$ .

(g) Use (b) to state Euler's identity.

**Exercise 1.3** (► ★ drill — rotation). Let  $z = 2 e^{i\pi/6}$ . Write down  $e^{i\pi/2} z$  in exponential form, and describe geometrically the effect of multiplying a complex number by (i)  $i$ , (ii)  $e^{i\pi}$ , (iii)  $e^{-i\pi/2}$ .

**Exercise 1.4** (★ drill — integer powers). Compute, using the exponential form:

(a)  $(1 + i)^4$ , (b)  $(\sqrt{3} + i)^3$ , (c)  $(1 - i)^6$ , (d)  $(2 e^{i\pi/5})^{10}$ .

(e) Use De Moivre's theorem to simplify  $(\cos \theta + i \sin \theta)^{-2}$ .

**Exercise 1.5** (★ drill — a few roots). Find all the (a) square roots of  $i$  and (b) fourth roots of  $1$ . Plot the roots in (b) on an Argand diagram, mark the unit circle, and name the polygon they form.

**Exercise 1.6** (★ drill — the fundamental theorem of algebra). (a) State the fundamental theorem of algebra, and say what “counted with multiplicity” means. (b) How many roots in  $\mathbb{C}$ , counted with multiplicity, has (i) a degree-5 polynomial; (ii)  $z^7 - 3 = 0$ ; (iii)  $(z - 1)^2(z^2 + 1) = 0$ ? (c) True or false: every real cubic has at least one real root?

**Exercise 1.7** (★ drill — hyperbolic link). Using  $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ , evaluate  $\cos(i)$  in terms of  $e$ . (The general identity  $\cos(ix) = \cosh x$  is built up in Exercise 2.9.)

**Exercise 1.8** (★ drill — phasors). (a) Write  $5 \cos(\omega t + \frac{\pi}{3})$  and  $2 \sin \omega t$  as phasors  $P = A e^{i\varphi}$ . (b) The phasor  $3 e^{-i\pi/2}$  represents which real signal  $A \cos(\omega t + \varphi)$ ?

## 2 Tutorial problems

**Exercise 2.1** (★★core — arithmetic in exponential form). Let  $z = \sqrt{3} + i$  and  $w = 1 + i$ . Write  $z$  and  $w$  in exponential form, then find  $zw$  and  $z/w$  in both exponential and Cartesian form.

**Exercise 2.2** (★★core — rotation in action). (a) Rotate  $z = 3 + 4i$  anticlockwise by  $\frac{\pi}{2}$  about the origin, and then (separately) clockwise by  $\frac{\pi}{2}$ . (b) Rotate  $2 + 2i$  anticlockwise by  $\frac{\pi}{4}$ . Give exact Cartesian answers.

**Exercise 2.3** (► ★★core — prove De Moivre (stretch)). Prove by induction that  $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$  for all integers  $n \geq 1$ . (Pure-maths.)

**Exercise 2.4** (► ★★core — powers by De Moivre (anchor)). Compute (a)  $(1 + i)^8$  and (b)  $(\sqrt{3} + i)^6$  using De Moivre's theorem.

**Exercise 2.5** (► ★★core — roots, two shapes (anchor)). Find (a) the cube roots of 8 and (b) the fourth roots of  $-16$ . In each case give exact Cartesian values and describe the polygon they form on the Argand diagram.

**Exercise 2.6** (► ★★core — sixth roots of unity (anchor)). Find all sixth roots of unity in Cartesian form, and verify that they sum to zero. Why does this generalise to all  $n \geq 2$ ?

**Exercise 2.7** (★★core — data-science flavour: twiddle factors). The fast Fourier transform is built from the “twiddle factors”  $W_N^k = e^{-i2\pi k/N}$ . Take  $N = 8$ . (a) Write  $W_8^1$  in Cartesian form. (b) Show that  $W_8^{k+4} = -W_8^k$  for every integer  $k$  (the symmetry that halves the FFT's work). (c) Give the modulus and argument of  $W_8^3$ .

**Exercise 2.8** (► ★★core —  $\cos 3\theta$  and  $\sin 3\theta$  (anchor)). Using De Moivre and the binomial theorem, show that  $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$  and  $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ .

**Exercise 2.9** (★★core — trig-hyperbolic link). (a) Show  $\cos(ix) = \cosh x$  and  $\sin(ix) = i \sinh x$ . (b) Hence evaluate  $\cos(i \ln 2)$  exactly.

**Exercise 2.10** (► ★★core — combining oscillations (anchor; physics)). (a) Combine  $3 \cos \omega t + 4 \sin \omega t$  into a single sinusoid  $R \cos(\omega t - \alpha)$ . (b) Combine  $2 \cos \omega t + 2 \cos(\omega t + \frac{\pi}{2})$  likewise.

## 3 Further practice (home)

### Roots and the fundamental theorem (Skills 5–6)

**Exercise 3.1** (★★core — solving  $w^n = z$  with a non-real right-hand side). (a) Solve  $w^3 = -27i$ , giving the three roots in exponential and Cartesian form. (b) Solve  $w^3 = 1 + i$ , leaving the roots in exponential form, and check that each root is the previous one rotated through  $\frac{2\pi}{3}$  about the origin.

**Exercise 3.2** (★★core — square root the algebraic way). The argument of  $5 - 12i$  is not a standard angle, so the exponential method is awkward. Instead find both square roots of  $5 - 12i$  in Cartesian form by equating real and imaginary parts of  $(a + ib)^2$ , and verify by squaring.

**Exercise 3.3** (★★core — FTA and real factors). (a) Verify that  $z = 2i$  is a root of  $z^2 + 4$ , and write down the other. (b) Factor  $z^4 + 4$  into two real quadratics. [Hint: find the four roots first.]

**Exercise 3.4** (★★★challenge — a conjugate equation). Find all  $z \in \mathbb{C}$  with  $z^2 = \bar{z}$ .

### Trigonometric identities (Skill 7)

**Exercise 3.5** (★★core — power reduction). Use the inverse Euler formula to write  $\sin^4 \theta$  as a sum of cosines of multiples of  $\theta$ .

**Exercise 3.6** (★★★challenge — a finite cosine sum). Using the geometric series for  $\sum_{k=0}^{n-1} e^{ik\theta}$  (with  $\theta \neq 0 \bmod 2\pi$ ), show that

$$\sum_{k=0}^{n-1} \cos k\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\left(\frac{(n-1)\theta}{2}\right).$$

### Trigonometric–hyperbolic link (Skill 8)

**Exercise 3.7** (★★core — the converse pair). Show that  $\cosh(ix) = \cos x$  and  $\sinh(ix) = i \sin x$ , and evaluate  $\sin(2i)$  and  $\cos(\pi i)$  in terms of  $e$ .

### Phasors (Skill 9)

**Exercise 3.8** (★★core — physics word problem: two AC sources). Two alternating voltages of equal amplitude but different phase,  $V_1 = 10 \cos \omega t$  and  $V_2 = 10 \cos(\omega t + \frac{2\pi}{3})$  (volts), are added. Find the amplitude and phase of  $V_1 + V_2$ .